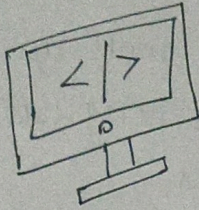


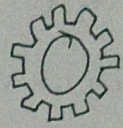
CONCEPT MAPPING

Q12: CONCEPT MAP- Algorithm $\rightarrow$ Best case, Average case, worst case $\rightarrow$ Time complexity $\rightarrow$ Performance evaluation



1. Algorithm

A finite set of instructions to solve a problem



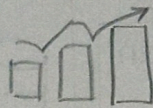
2. Best case



Best possible Input Instance

Minimum running time

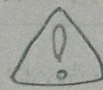
3. Average-Case



Average over all possible inputs

Expected running time

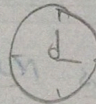
Worst Case



Worst possible input Instance

Maximum running time

5. Time complexity



Describes how running time grows with input size n



6. Asymptotic Notation

Big O (O)

Upper Bound

$$T(n) = O(g(n))$$

Gives the maximum growth rate

Big Ω (Omega)

Lower bound

$$T(n) = \Omega(g(n))$$

Gives the minimum growth rate

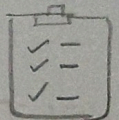
Big Θ (Theta)

Tight Bound

$$T(n) = \Theta(g(n))$$

Gives both upper & lower bounds

7. Performance evaluation



used to compare algorithms & choose the most efficient one

Q28 CONCEPT MAP: Algorithm $\rightarrow$ Efficiency $\rightarrow$ Time complexity & Space complexity $\rightarrow$ Trade-off:

